

FAYETTEVILLE, NC, USA

WMO: 723035

Lat: 34.991N

Lon: 78.880W

Elev: 186

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93740

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.2	(c) 26.4	(d) 2.8	(e) 6.4	(f) 30.3	(g) 7.8	(h) 8.3	(i) 34.2	(j) 24.9	(k) 54.5	(l) 21.6	(m) 53.8	(n) 6.3	(o) 30	(p) 0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.7	(c) 96.4	(d) 76.1	(e) 93.7	(f) 75.2	(g) 91.5	(h) 74.7	(i) 79.1	(j) 89.9	(k) 78.0	(l) 88.4	(m) 77.2	(n) 87.0	(o) 9.6	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	138.4	82.5	75.3	133.3	81.6	74.5	130.1	81.0	42.6	90.0	41.5	88.8	40.6	87.2	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.4	(b) 18.0	(c) 16.0	(e) 15.0	(f) 100.4	(g) 4.9	(h) 2.6	(i) 11.5	(j) 102.2	(k) 8.6	(l) 103.7	(m) 5.8	(n) 105.1	(o) 2.2	(p) 107.0		
(4) 20.4	(5) 18.0	(6) 16.0	(7) 13.0	(8) 81.2	(9) 4.7	(10) 2.0	(11) 9.6	(12) 82.6	(13) 6.9	(14) 83.8	(15) 4.2	(16) 84.9	(17) 0.8	(18) 86.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	44.2	47.5	54.2	63.3	71.2	78.5	81.4	80.0	74.4	64.2	53.7	47.4		
	DBStd	15.19	10.49	9.57	9.27	7.90	6.64	4.97	3.82	4.13	5.49	7.68	8.52	9.37		
	HDD50	685	243	148	65	5	0	0	0	0	0	2	57	165		
	HDD65	2659	647	494	353	124	23	1	0	0	5	110	351	551		
	CDD50	5583	63	78	193	404	658	856	974	931	733	442	167	84		
	CDD65	2083	2	4	17	74	216	407	509	466	288	84	11	5		
	CDH74	19683	5	30	171	725	2001	4022	5276	4494	2245	633	65	16		
	CDH80	8338	0	2	25	190	748	1855	2503	2034	824	152	5	0		
(14) Wind	WSAvg	7.2	7.6	7.9	8.5	8.8	7.7	6.9	6.5	5.9	6.6	6.6	6.8	6.9		
(15) Precipitation	PrecAvg	46.8	3.8	3.5	4.0	3.0	3.3	4.4	5.7	5.4	4.5	3.0	2.9	3.1		
	PrecMax	65.8	7.8	6.6	8.9	7.2	10.0	15.9	14.6	10.7	14.7	9.3	6.4	7.7		
	PrecMin	32.4	0.8	0.6	0.3	0.2	0.6	0.4	0.6	1.1	0.0	0.5	0.3	0.5		
	PrecStd	7.1	1.7	1.7	1.8	1.6	1.8	2.7	2.7	2.5	3.2	2.1	1.7	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.9	78.8	83.1	88.9	94.6	98.6	99.2	99.2	93.6	88.6	80.0	76.2		
		MCWB	63.9	64.4	64.1	68.4	72.1	75.7	77.8	76.4	72.8	72.5	67.0	65.7		
	2%	DB	69.9	72.7	78.9	84.8	90.6	94.9	96.2	95.4	90.6	84.0	75.3	71.8		
		MCWB	61.0	60.3	62.8	66.2	71.0	74.9	77.0	76.2	73.4	70.1	63.8	64.1		
	5%	DB	65.6	68.3	74.8	81.8	87.7	92.4	93.4	92.5	88.2	80.8	72.3	67.6		
		MCWB	58.7	57.9	60.1	64.6	70.1	74.1	76.2	75.7	73.0	67.7	62.8	62.1		
	10%	DB	61.5	64.0	70.3	78.6	84.1	89.9	91.1	90.0	85.0	77.2	68.5	63.4		
		MCWB	55.3	55.3	58.1	63.2	68.9	73.4	75.8	75.3	71.9	67.3	60.9	57.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.0	67.2	68.3	72.3	75.3	78.9	81.2	80.2	78.1	76.4	70.9	69.1		
		MCDB	70.7	74.5	76.4	80.7	86.3	90.5	92.7	90.8	87.1	80.4	74.1	73.2		
	2%	WB	63.5	64.0	65.8	69.8	73.7	77.5	79.4	78.7	76.5	73.7	67.7	65.9		
		MCDB	67.4	69.7	73.7	79.0	84.3	88.9	90.6	89.1	84.7	79.9	72.1	69.9		
	5%	WB	60.1	60.4	63.7	68.0	72.4	76.4	78.3	77.8	75.4	71.6	65.4	62.7		
		MCDB	64.4	66.4	70.0	77.1	82.6	87.1	89.0	87.7	82.9	77.2	70.2	66.6		
	10%	WB	56.1	56.7	61.4	66.0	71.0	75.4	77.4	76.9	74.3	69.2	62.5	58.7		
		MCDB	60.6	62.7	69.0	74.9	80.4	85.3	87.5	86.1	80.8	75.2	67.3	63.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	20.9	22.1	22.7	21.3	19.5	18.7	18.5	18.2	20.6	21.6	19.2		
		MCDBR	23.6	25.8	26.5	26.1	24.4	22.5	21.1	21.7	21.6	22.9	24.3	22.3		
		MCWBR	16.0	16.1	13.4	11.1	8.5	6.9	6.7	6.3	7.5	10.5	14.3	15.2		
	5% WB	MCDBR	20.7	23.3	22.1	21.4	20.1	19.5	18.8	18.7	17.1	16.9	19.5	20.1		
		MCWBR	16.6	17.5	14.2	11.1	8.2	7.1	6.8	6.4	7.0	9.2	14.2	16.0		
(40) Clear-Sky Solar Irradiance	taub		0.322	0.334	0.370	0.408	0.466	0.501	0.548	0.524	0.432	0.392	0.343	0.319		
	taud		2.481	2.439	2.373	2.264	2.146	2.090	1.979	2.035	2.290	2.367	2.463	2.523		
	Ebn at Noon		280	288	286	279	263	252	240	243	262	264	269	274		
	Edh at Noon		27	31	36	43	49	51	57	53	39	33	27	24		
(44) All-Sky Solar Radiation	RadAvg		845	1055	1395	1750	1919	2021	1923	1731	1467	1206	928	738		
	RadStd		52	115	103	118	149	134	119	89	120	150	75	61		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+304	+227
Regional (65 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+283	+203

Nomenclature: See separate page